Syed Ibrahim Omer

ibrahsyed2-c@my.cityu.edu.hk | (+852)44589107 | Homepage | Linkedin | Google Scholar

Education

City University of Hong Kong Expected May 2028

Bachelor of Science (BS) in Data Science

Distinctions: Dean's List, Full Academic Scholarship, Tiger Programme Member, Diversity Grant Recipient

Relevant Coursework: Multi-variable Calculus, Linear Algebra, Advanced Statistics, Bayesian Analysis, Convex Optimization, Introduction to Finance, Introduction to Economics

Skills

Languages: Python, C/C++, CUDA, Rust, SQL, Lean 4

Data: Polars, DuckDB, Impala, Kafka, Parquet/Avro

ML: PyTorch, scikit-learn, LightGBM, HuggingFace

Infra: Linux, Docker, K8s, Git, W&B, Nsight

Experience

NVIDIA AI Technology Center (NVAITC) Hong Kong SAR

Student Ambassador (Research Track) *Jun 2025 – Present*

- First author of **MaRK** (accepted to NeurIPS 2026 Main Track Conference): conditions a frozen 111M-parameter model with **6–11M adapters**, cutting trainable parameters by 90%
- Co-authoring **BXINT4**: a new low-precision training datatype with **3.3**× faster 4-bit GEMM than FP16 on GPUs without native 4-bit tensor cores (Turing, Ampere, and Ada GPUs)
- Nominated as NVIDIA DLI workshop instructor for "Rapid Application Development with Large Language Models" course, training engineers on cutting-edge developments in AI and applying them in applications

Habib Bank Limited (HBL) Karachi, Pakistan

Machine Learning Intern, Data Lake Function *Jun 2026 – Aug 2026*

- Built an end-to-end, air-gapped churn pipeline (DuckDB ETL → LightGBM → reporting) answering **who** churns, **when**, and **why** for **32k** bank accounts
- Reached **0.93 weighted F1** at account level by racing a joint multiclass model against a discrete-time hazard model, replacing branch reports that only described churn after it happened
- Converted TreeSHAP into per-account risk drivers and 12-month what-if hazard curves, giving retention teams a ranked priority list and a peak-risk month to time offers

Publications

MaRK: Markov-adapted Recurrent Kernels | *First Author* NeurIPS 2026

- Designed the conditioning mechanism and **3 adapter variants** (hypernet, Chebyshev, DCT) over Mamba-style state-space layers, proving layer-wise stability in **Lean 4**
- Derived an $O(N^{-1/2})$ operator-recovery rate and validated it on **5 datasets** × **10 seeds** against synthetic ground truth; fixed a kernel limit that blocked mixed-mask batching

Redefining Hybrid Blockchains | *Sole Author* arXiv:2504.18966

- Cited in peer-reviewed work: $\mathcal{O}(N^2) \to \mathcal{O}(N)$ consensus traffic at **1,030 TPS** and 2.2 s median finality

Projects

Markowitz Portfolio Engine | *optimization, risk* GitHub

- Built a **60/40** fixed-income and growth portfolio on Markowitz mean-variance weights and **20,000 Monte Carlo paths**: 18.3% backtested return at 11.5% volatility, ∼60% above the S&P 500
- Hedged with collar positions and correlation screening; **70% modeled odds** of beating the S&P 500 over 10Y

News-Sentiment Trading Dashboard | *NLP, backtesting* GitHub

- Built a **Streamlit** dashboard over 10 years of prices and **7,000+** news articles, scoring headlines daily with FinBERT and VADER and clustering them with K-Means
- Tested whether sentiment leads price using lagged correlations with p-values, then backtested buy/sell, dollar-cost-averaging and lump-sum strategies on the signals

Migration Surge Forecasting | *lead-lag time series* GitHub

- Modeled US migration surges for 15 countries from visa, border-encounter, FX and search-interest series; FX leads visas by **2–6 months** ($p < 0.05$), giving advance warning for planning
- Showed flows are **Zipfian**: the top 20% of countries hold **88.3%** of all visas, so forecasts target few corridors

Open Source

Market Indicators Toolkit | *indicators-cli on PyPI* PyPI

- Published a CLI computing **9 indicators** (MACD, RSI, Bollinger, ATR, OBV, ROC, Stochastic) in **Polars**
- Shipped CPU and GPU engines, batch tickers, **5 export formats** and 27 functional tests on live market data